Preliminaries

Contents

1 Introduction

Before we delve into the laws of thermodynamics, it is essential to establish the fundamental concepts, definitions, and terminology that underpin thermodynamic reasoning. These preliminary ideas provide a common language for describing macroscopic systems, clarify the meaning of equilibrium and state variables, and set the stage for the logical development of the subject. A clear understanding of these foundations will be crucial for appreciating both the physical content and the mathematical structure of the laws that follow.

2 Thermodynamic State Variables

A thermodynamic system is completely specified by a set of measurable macroscopic quantities known as **state variables** (or **state functions**). These variables characterize the equilibrium state of the system and depend only on the current state, not on the particular process by which the state was prepared.

State variables are commonly classified into two categories:

- **Extensive variables** are proportional to the size or amount of substance in the system. If the system is scaled by a factor λ, these quantities scale in the same way. Typical examples include
  - E: internal energy,
  - S: entropy,
  - V: volume,
  - N: number of particles.
- **Intensive variables** are independent of the system size and remain unchanged under subdivision of the system. Representative examples are
  - T: temperature,
  - P: pressure,
  - μ: chemical potential.

The distinction between extensive and intensive variables plays a central role in thermodynamics, particularly in the construction of thermodynamic potentials, the formulation of equilibrium conditions, and the analysis of composite and interacting systems.

3 Equilibrium and Quasi-static Processes

- A system is in **thermodynamic equilibrium** when its macroscopic properties are stationary in time and spatially uniform, so that no spontaneous macroscopic flows occur. Thermodynamic equilibrium comprises the simultaneous fulfillment of:
  - *Thermal equilibrium*: uniform temperature throughout the system;
  - *Mechanical equilibrium*: absence of unbalanced forces or pressure gradients;
  - *Chemical equilibrium*: absence of net chemical reactions or particle transport between subsystems.
- A **quasi-static (or quasi-equilibrium) process** is an idealized process carried out infinitely slowly, so that the system evolves through a continuous succession of equilibrium states. While no real process is strictly quasi-static, this approximation is essential in thermodynamics, as it ensures that quantities such as pressure and temperature are well defined at each instant, permitting meaningful differential expressions for work and heat.

Example. A sufficiently slow compression of a gas in a piston, during which pressure and temperature remain well defined and nearly uniform throughout the system, may be treated as a quasi-static process.

4 Thermodynamic System and Surroundings

A **thermodynamic system** is the part of the universe selected for analysis. All regions external to the system constitute the **surroundings** (or **environment**). The separation between the system and its surroundings is defined by a **boundary**, which may be real or imaginary, fixed or movable.

Depending on the nature of this boundary, thermodynamic systems are classified as follows:

- **Open system**: exchanges both energy and matter with its surroundings;
- **Closed system**: exchanges energy but not matter;
- **Isolated system**: exchanges neither energy nor matter.

The choice of system and its boundary is dictated by the physical situation under consideration and determines which exchanges are permitted and which thermodynamic variables may vary.

5 Notation Conventions

In thermodynamics, it is essential to distinguish between *exact* and *inexact* differentials, reflecting the fundamental difference between state functions and process-dependent quantities:

- dE, dS, dV, dN: These denote **exact differentials** associated with thermodynamic state functions. Their integrals depend only on the initial and final equilibrium states of the system and are independent of the path connecting them.
- δQ, δW: These denote **inexact differentials** associated with heat and work, respectively. Heat and work are not properties of the system but of the thermodynamic process, and their values depend on the path taken between states.
- **Sign convention:** Throughout these notes, heat absorbed by the system and work performed by the system on its surroundings are taken to be positive. With this convention, the first law of thermodynamics is written as

  $$\delta Q = dU + \delta W,$$

  where $\delta Q > 0$ corresponds to heat flowing into the system and $\delta W > 0$ corresponds to work done by the system on the external environment.

6 Thermodynamic Description and Scale

Thermodynamics is a **macroscopic theory** that describes physical systems in terms of measurable bulk properties such as pressure, temperature, energy, entropy, and volume.

These quantities are defined and experimentally accessible without reference to microscopic details, such as the positions and velocities of individual particles.

A defining feature of thermodynamics is that it concerns the *collective behavior* of systems containing an enormous number of microscopic degrees of freedom. For example, one mole of an ideal gas contains approximately Avogadro's number of particles,

$$N_A \approx 6.022 \times 10^{23}.$$

Despite the enormous number of microscopic degrees of freedom, macroscopic observables remain well defined, stable, and reproducible. Microscopic randomness gives rise only to small fluctuations, whose relative magnitude diminishes with increasing system size in accordance with the Law of Large Numbers (see Appendix A).

This macroscopic perspective is both a strength and a limitation. Thermodynamics provides powerful and universal predictions for equilibrium properties and state changes—such as equations of state and phase transitions—without requiring detailed microscopic models. At the same time, it does not explain the microscopic origin of thermodynamic quantities or account for fluctuations, which must be addressed using statistical mechanics.

Key Scales in Thermodynamics

The applicability of a thermodynamic description depends on the scale at which a system is examined:

- **Microscopic scale:** Characterized by atomic or molecular lengths (typically nanometers), energies on the order of k_BT, and short time scales such as collision times.
- **Macroscopic scale:** Characterized by lengths large compared with microscopic scales (e.g., millimeters or larger), bulk thermodynamic variables, and time scales over which equilibrium is established.
- **Mesoscopic scale:** Intermediate between microscopic and macroscopic, where fluctuations may be significant and neither a purely thermodynamic nor a fully microscopic description is sufficient. Examples include nanoscale devices and biological systems such as cells.

Thermodynamics is most accurate and predictive at the macroscopic scale, where fluctuations are negligible and systems can be meaningfully described as being in, or close to, equilibrium.

7 Statistical Mechanics: A Microscopic Foundation

Statistical mechanics provides the microscopic foundation of thermodynamics by connecting the underlying dynamics of many-particle systems—governed by classical or quantum mechanics—to macroscopic thermodynamic laws. Within this framework, thermodynamic quantities arise naturally as *statistical averages* over large ensembles of microscopic states.

A microscopic state, or *microstate*, is specified by the complete set of dynamical variables describing the system: the positions and momenta of all particles in classical mechanics, or the quantum state (wavefunction or density operator) in quantum mechanics. For systems containing on the order of 10^{23} particles, it is neither feasible nor meaningful to track the exact microstate. Instead, statistical mechanics describes the system in terms of a probability distribution over all microstates consistent with specified macroscopic constraints, such as fixed energy, volume, and particle number.

Macroscopic observables—including internal energy, pressure, entropy, and particle number—are then obtained as *ensemble averages* with respect to this probability distribution. In doing so, statistical mechanics not only reproduces the laws of thermodynamics but also explains their microscopic origin, providing insight into entropy, fluctuations, irreversibility, and phase transitions.

Appendices

A The Law of Large Numbers

A fundamental mathematical principle underlying the macroscopic description of matter is the *law of large numbers.* This theorem explains why thermodynamic quantities, which are built from microscopic degrees of freedom that exhibit random behavior, nonetheless appear stable, reproducible, and effectively deterministic at the macroscopic level.

In statistical mechanics, macroscopic observables are typically expressed as averages over a very large number of microscopic quantities. Examples include the mean kinetic energy of particles, which determines the temperature, or the momentum transfer per unit area, which gives rise to pressure. The law of large numbers provides the mathematical justification for replacing such microscopic averages by their expected values when the number of particles is large.

From a probabilistic standpoint, a microscopic quantity is modeled as a random variable. The *expected value* of a random variable X, denoted by

$$\mu = \mathbb{E}[X],$$

is defined as the probability-weighted average of all possible values of X. Physically, μ represents the ensemble average of the corresponding microscopic observable.

When a system contains N particles, one may consider N independent and identically distributed random variables $X_1, X_2, \ldots, X_N$, each representing the same microscopic quantity for a different particle. The corresponding macroscopic observable is then given by the sample mean

$$\overline{X}_N = \frac{1}{N} \sum_{i=1}^{N} X_i.$$

The law of large numbers states that, provided the expected value μ is finite, the sample mean $\overline{X}_N$ converges to μ as N becomes large. In practical terms, this means that the relative fluctuations of $\overline{X}_N$ decrease as the system size increases, typically scaling as $N^{-1/2}$. For macroscopic systems with $N \sim 10^{23}$ particles, these fluctuations are negligibly small.

As a result, macroscopic quantities such as temperature, pressure, and density can be treated as well-defined and reproducible properties of a system, despite the randomness of the underlying microscopic dynamics. This principle forms a cornerstone of statistical mechanics and provides the mathematical basis for the macroscopic laws of thermodynamics.